

Instructions to the Students:

1. *All the questions are compulsory.*
 2. *Use of non-programmable scientific calculators and steam table is allowed.*
 3. *Assume suitable data wherever necessary and mention it clearly.*
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Q. 1 Solve Any Two of the following. (6 x 2) 12 M

- A) Derive the combustion equation for hydrocarbon (solid) fuels
- B) Explain the working of La-Mont Boiler giving its principle and schematic sketch.
- C) Define Calorific value and distinguish between gross calorific value and net calorific value.

Q.2 Solve Any Two of the following. (6 x 2) 12 M

- A) List any two Accessories and Mountings. Explain Economizer in details with the help of neat sketch.
- B) Describe with a neat diagram, the construction and working of a Babcock and Wilcox water tube boiler.
- C) Give the detailed classification of draught. Also explain with neat sketch Induced mechanical draught.

Q. 3 Solve Any Two of the following. (6 x 2) 12 M

- A) Explain ideal Rankine cycle on P-V, T-S & h-s diagram also derive the expression for thermal efficiency of cycle.
- B) Classify steam turbine. Differentiate between impulse and reaction steam turbines
- C) Define steam condenser. Explain surface condenser in detail.

Q.4 Solve Any Two of the following. (6 x 2) 12 M

- A) What is use of cooling tower used in steam power plant? Explain Induced draught cooling tower with neat sketch.
- B) Define nozzle; Also derive expression for condition for maximum discharge through the steam nozzle.
- C) A single acting reciprocating air compressor has cylinder diameter and stroke of 200 mm and 300 mm respectively. The compressor sucks air at 1 bar and 27⁰ C and delivers at 8 bars while running at 100 r.p.m. Find: a. Indicated power of compressor; b. Mass of air delivered by the compressor per minute; c. Temperature of the air delivered by the compressor.

Q. 5 Solve Any Two of the following.

(6 x 2) 12 M

- A) Draw and explain in short P-V diagram of single stage Reciprocating compressor with and without clearance
- B) The following observations were made on a boiler plant during one hour test:
Steam pressure = 20 bar; Steam temperature = 260°C ; Steam generated = 37500 kg;
Temperature of water entering the economizer = 15°C ; Temperature of water leaving the economizer = 90°C ; Fuel used = 4400 kg; Energy of combustion of fuel = 30 000 kJ/kg.
Calculate:
a. The equivalent evaporation per kg;
b. The thermal efficiency of the plant;
c. The percentage heat energy of the fuel energy utilized by the economizer.
- C) With the help of neat sketch explain the working of centrifugal compressor. Also write down the formula to calculate the work done by centrifugal compressor for different processes.

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